

ASCHAM SCHOOL
MATHEMATICS EXTENSION 2
TRIAL EXAMINATION

2003

Time : 3 hours + 5 minutes reading time

Instructions:

Attempt all questions

All questions are of equal value

All necessary working should be shown for every question.

Full marks may not be awarded for careless or badly arranged work

A Table of Standard Integrals is provided

Approved calculators may be used

Each question should be answered in a separate booklet

Question 1

- a) $(2-3i)(4+i) = p+iq$ where $p, q \in \mathbb{R}$. Find p and q . [1]
- b) (i) Express $z = -\sqrt{3} + i$ in modulus-argument form. [2]
- (ii) Hence show that $z^7 + 64z = 0$ [2]
- c) Sketch the following subsets of the Argand diagram, showing important features and intercepts with the axes.
- (i) $\{z : 1 < |z| \leq 3 \text{ and } 0 < \arg z \leq \frac{\pi}{2}\}$ [2]
- (ii) $\{z : |z+1| + |z-1| = 3\}$ [3]
- (iii) $\{z : \arg(z-2) - \arg(z+2) = \frac{\pi}{3}\}$ [2]
- d) Find the Cartesian form of the equation of the locus of the point z if $\operatorname{Re}\left[\frac{z-4}{z}\right] = 0$ [3]

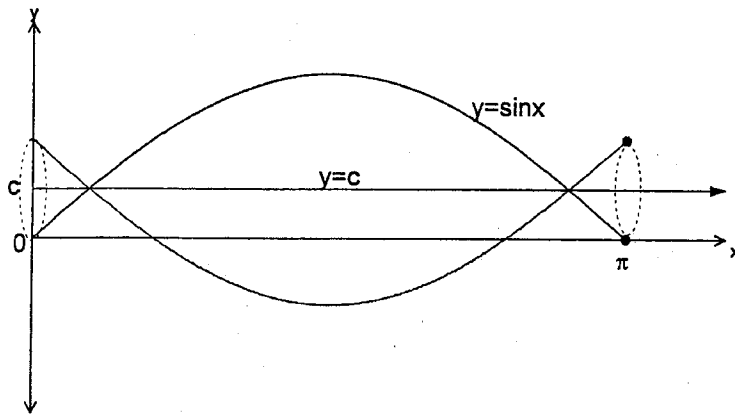
Question 2 *Please take a new booklet*

- a) Find $\int \frac{e^{2x}}{e^x + 1} dx$ [2]
- b) Evaluate $\int \tan^3 x \, dx$ [2]
- c) Evaluate $\int_0^{\pi} e^x \sin x \, dx$ [3]
- d) (i) Show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{x(\pi-2x)} = \frac{2}{\pi} \ln 2$ [3]
- (ii) Using the substitution $u = a+b-x$, show that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ [2]
- (iii) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx$ [3]

Question 3 Please take a new booklet

- a) A chocolate has a circular base of radius 1 cm. If every section perpendicular to this base is an equilateral triangle, find the volume of chocolate needed to make a box of 40 such chocolates. [6]

b)



The arch $y = \sin x$, $0 \leq x \leq \pi$, is revolved around the line $y = c$ to generate the solid shown.

- (i) Show that the volume generated is given by $\pi(\pi c^2 - 4c + \frac{\pi}{2})$ [6]
- (ii) Find the value of c which minimises the volume. [3]

Question 4 Please take a new booklet

- a) A ball of mass m is thrown vertically upwards under gravity, the air resistance to the motion being $\frac{mgv^2}{a^2}$ where the speed is v , a is a constant and g is the acceleration due to gravity.

- (i) Show that during the upward motion of the ball

$$v \frac{dv}{dx} = \frac{-g}{a^2} (a^2 + v^2)$$

where x is the upward displacement. [2]

- (ii) Show that the greatest height reached is $\frac{a^2}{2g} \ln \left(1 + \frac{u^2}{a^2} \right)$ where u is the speed of projection. [5]

- b) A curve is defined by the parametric equations $x = \cos^3 \theta$, $y = \sin^3 \theta$ for $0 < \theta < \frac{\pi}{4}$.
- (i) Show that the equation to the normal to the curve at the point $P (\cos^3 \phi, \sin^3 \phi)$ is $x \cos \phi - y \sin \phi = \cos 2\phi$ [4]
- (ii) The normal at P cuts the x -axis at A and the y -axis at B . Show $AB = 2 \cot 2\phi$ [4]

Question 5 *Please take a new booklet*

- a) If $ax^3 + bx^2 + d = 0$ has a double root, show that $27a^2d + 4b^3 = 0$ [3]
- b) (i) Prove that $P(x) = \frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x + c$ has no real zeros if $c > 9\frac{1}{3}$ [4]
- (ii) Explain why the largest zero of $P(x)$ is greater than 2 if $c = -2$. Find an approximation for the largest zero of $P(x)$ using one application of Newton's method. [3]
- c) (i) P is any point inside a circle center O . M is the midpoint of chords AB through P . Find the locus of M . Explain your answer. [3]
- (ii) Q is any point outside a circle center C . N is the midpoint of chords DE through Q . State the locus of N . [2]

Continued on next page

Question 6 Please take a new booklet

a) (i) Find the five fifth roots of unity. [2]

(ii) If $\omega = \text{cis } \frac{2\pi}{5}$, show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ [3]

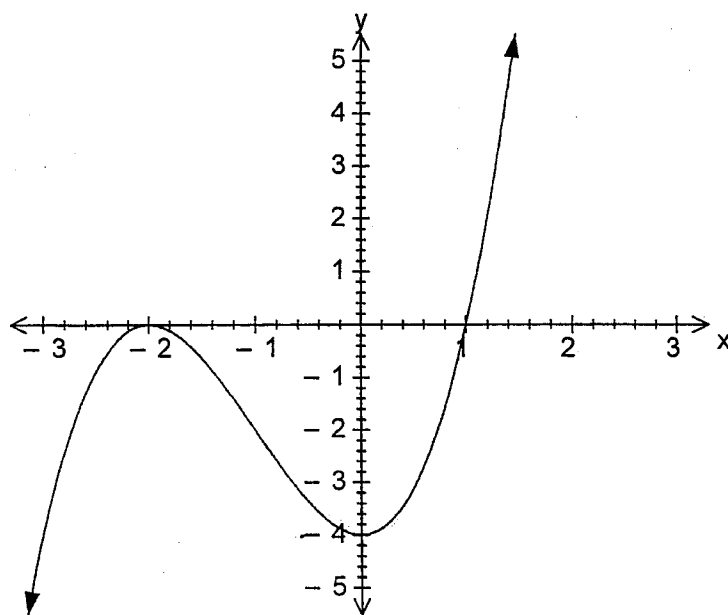
(iii) Show that $z_1 = \omega + \omega^4$ and $z_2 = \omega^2 + \omega^3$ are roots of the equation $z^2 + z - 1 = 0$ [3]

b) (i) By using the expansions of $\cos(x-y)$ and $\cos(x+y)$ show that $\sin x \sin y = \frac{1}{2}(\cos P - \cos Q)$ where $P = (x-y)$ and $Q = (x+y)$ [3]

(ii) Hence prove that $\sin x + \sin 3x + \sin 5x + \dots + \sin(2n-1)x = \frac{\sin^2 nx}{\sin x}$ [4]

Question 7 Please take a new booklet

a) The graph of $y = x^3 + 3x^2 - 4$ is sketched below



(i) Sketch the curves $y = |x^3 + 3x^2 - 4|$ and $y = \ln|x^3 + 3x^2 - 4|$ on separate axes. [3]

(ii) Hence or otherwise determine the value of m , where m is a constant, such that the equation $2\ln|x+2| + \ln|x-1| = m$ [4]

- b) AB is a diameter of a circle whose centre is O and C is a point on the circumference such that $\angle AOC$ is acute. OC is produced to meet the tangent at A in D. Let $\angle CBD = \alpha$ and $\angle ABC = \beta$. Prove

(i) $\tan(\alpha + \beta) = \frac{1}{2} \tan 2\beta$ [3]

(ii) $\tan \alpha = \tan^3 \beta$ [3]

(iii) Calculate the value of α when $AD = AB$ [2]

Question 8 *Please take a new booklet*

- a) (i) Show that the condition for the line $y = mx + c$ to be tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 + b^2$ [3]
- (iii) Hence or otherwise prove that the pair of tangents from the point (3, 4) to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to each other. [4]

b) Let $I_{2n} = \int_1^1 (1-x^2)^n dx$ where $n \geq 0$

(i) Use the substitution $x = \sin \theta$ to show that $I_{2n} = \frac{2n}{2n+1} I_{2n-2}$ [3]

(ii) Show that $I_6 = \frac{32}{35}$ [2]

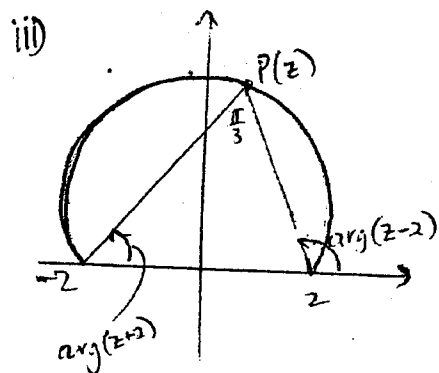
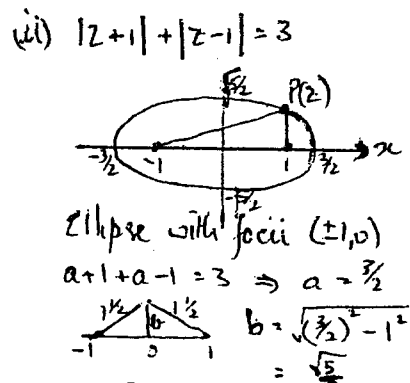
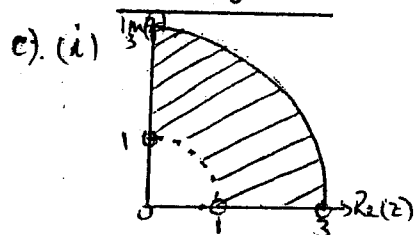
(iii) Deduce that $I_{2n} = \frac{2^{2n+1} (n!)^2}{(2n+1)!}$ [3]

End of Examination

1a) $(2-3i)(4+i)$
 $= 8-10i-3i^2$
 $= 11-10i$
 $p=11, q=-10$

b) i) $z = \sqrt{3} + i$
 $|z| = \sqrt{3+1}$
 $= 2$
 $\arg z = \tan^{-1}(-\frac{1}{\sqrt{3}})$
 $= \frac{5\pi}{6}$

$z = 2 \operatorname{cis} \frac{5\pi}{6}$
 ii) $z^7 + 64z = 2^7 (\operatorname{cis} \frac{5\pi}{6})^7 + 64 \times 2 \operatorname{cis} \frac{5\pi}{6}$
 $= 128 \operatorname{cis} \frac{35\pi}{6} + 128 \operatorname{cis} \frac{5\pi}{6}$
 $= 128 (\operatorname{cis} (-\frac{\pi}{6}) + \operatorname{cis} \frac{5\pi}{6})$
 $= 128 (\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} - \cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 $= 0$



d) $\operatorname{Re}(\frac{z-4}{z}) = 0$
 let $z = x+iy$
 $\frac{z-4}{z} = \frac{x+iy-4}{x+iy} \times \frac{x-iy}{x-iy}$
 $= \frac{x^2+y^2-4x+4iy}{x^2+y^2}$
 $\operatorname{Re}(\frac{z-4}{z}) = 1 - \frac{4x}{x^2+y^2} = 0$
 $x^2+y^2 = 4x$

2a) $\int \frac{e^{2x}}{e^x+1} dx = \int e^x - \frac{e^x}{e^x+1} dx$
 $= e^x - \ln(e^x+1) + C$

b) $\int \tan^3 x dx = \int \tan x (\sec^2 x - 1) dx$
 $= \int \tan x \sec^2 x - \frac{\sin x}{\cos x} dx$
 $= \frac{1}{2} \tan^2 x + \ln |\cos x| + C$

c) $\int_0^{\pi/2} e^x \sin x dx = I$
 $\begin{cases} u = \sin x \\ du = \cos x dx \\ dv = e^x dx \\ v = e^x \end{cases}$
 $I = [e^x \sin x]_0^{\pi/2} - \int_0^{\pi/2} e^x \cos x dx$
 $= 0 - \int_0^{\pi/2} e^x \cos x dx$
 $u = \cos x$
 $du = -\sin x$
 $dv = e^x dx$
 $v = e^x$
 $= -[e^x \cos x]_0^{\pi/2} - \int_0^{\pi/2} e^x \sin x dx$
 $= -[e^{\pi/2}(-1) - 1] - I$

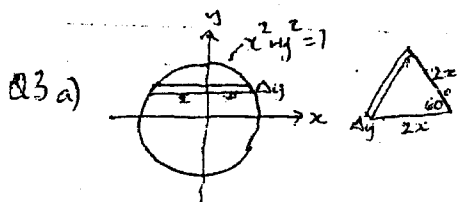
Ascham 2003 Ext 2 Trial. (1)

d) $\frac{1}{x(\pi-2x)} = \frac{A}{x} + \frac{B}{\pi-2x}$
 (i) $1 = A(\pi-2x) + Bx$
 $A = \frac{1}{\pi} \quad B = \frac{2}{\pi}$
 $\int_{\pi/6}^{\pi/3} \frac{dx}{x(\pi-2x)} = \int_{\pi/6}^{\pi/3} \frac{\frac{1}{\pi}}{x} + \frac{\frac{2}{\pi}}{\pi-2x} dx$
 $= \frac{1}{\pi} [\ln x - \ln(\pi-2x)]_{\pi/6}^{\pi/3}$
 $= \frac{1}{\pi} \left[\ln \frac{x}{\pi-2x} \right]_{\pi/6}^{\pi/3}$
 $= \frac{1}{\pi} \left[0 - \ln \frac{1}{4} \right]$
 $= \frac{2}{\pi} \ln 2$

(ii) Let $u = a+b-x$ $x=a \quad x=b$
 $du = -dx$ $u=b \quad u=a$
 $\int_a^b f(a+b-x) dx = \int_b^a f(u) \cdot (-du)$
 $= \int_a^b f(u) du$
 $= \int_a^b f(x) dx$

(iii) $\int_{\pi/6}^{\pi/3} \frac{\cos^2 x}{x(\pi-2x)} dx = I$
 $I = \int_{\pi/6}^{\pi/3} \frac{\cos^2(\frac{\pi}{6} + \frac{\pi}{3} - x)}{(\frac{\pi}{6} + \frac{\pi}{3} - x)(\pi - 2(\frac{\pi}{6} + \frac{\pi}{3} - x))} dx$
 $= \int_{\pi/6}^{\pi/3} \frac{\sin^2 x}{(\frac{\pi}{2} - x)(2x)} dx$
 $= \int_{\pi/6}^{\pi/3} \frac{1 - \cos^2 x}{(\pi-2x)x} dx$
 $= \int_{\pi/6}^{\pi/3} \frac{1}{x} - \frac{\cos^2 x}{x} dx$

$$I = \frac{1}{\pi} \ln 2$$



$$\Delta V = \frac{1}{2} \cdot 2x \cdot 2x \sin 60^\circ \Delta y$$

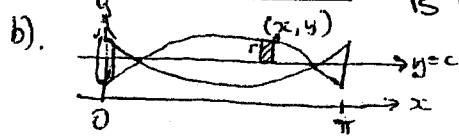
$$= dx^2 \cdot \frac{\sqrt{3}}{2} \Delta y$$

$$V = \int_{-1}^1 \sqrt{3} (1-y^2) dy$$

$$= 2\sqrt{3} \int_0^1 (1-y^2) dy$$

$$= 2\sqrt{3} \left[y - \frac{y^3}{3} \right]_0^1$$

$$= \frac{4}{3} \sqrt{3} \text{ cm}^3 \therefore \text{Total vol of choc is } \frac{16\sqrt{3}}{3} \text{ cm}^3$$



(i) $r = y - c$

$$\Delta V = \pi r^2 \Delta x$$

$$= \pi (y-c)^2 \Delta x$$

$$V = \pi \int_0^\pi (\sin x - c)^2 dx$$

$$= \pi \int_0^\pi \sin^2 x - 2c \sin x + c^2 dx$$

$$= \pi \int_0^\pi \frac{1}{2} (1 - \cos 2x) - 2c \sin x + c^2 dx$$

$$= \pi \left[\frac{x}{2} - \frac{1}{4} \sin 2x + 2c \cos x + c^2 x \right]_0^\pi$$

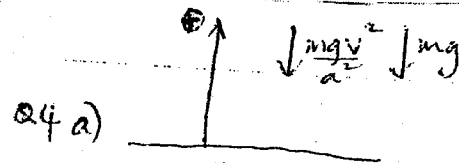
(ii) $\text{Min occurs when } v' = 0, v'' > 0$
 $v' = \pi [0 - 4 + 2c\pi] = 0$

$$c = \frac{4}{2\pi}$$

$$= \frac{2}{\pi}$$

$$v'' = 2\pi^2 > 0$$

$$\therefore \text{Minimum occurs when } c = \frac{2}{\pi}$$



$$F = m\ddot{x}$$

(i) $m\ddot{x} = -\frac{mgv^2}{a^2} - mg$

$$\ddot{x} = -\frac{g}{a^2} v^2 - g$$

$$\frac{dv}{dt} = -\frac{g}{a^2} (v^2 + a^2)$$

$$\frac{dv}{dx} \cdot \frac{dx}{dt} = -\frac{g}{a^2} (a^2 + v^2)$$

$$v \cdot \frac{dv}{dx} = -\frac{g}{a^2} (a^2 + v^2)$$

(ii) $\frac{dv}{dx} = -\frac{g}{a^2} \left(\frac{a^2}{v} + v \right)$

$$\frac{dx}{dv} = -\frac{a^2}{g} \left(\frac{v}{a^2 + v^2} \right)$$

$$x = -\frac{a^2}{g} \int \frac{v}{a^2 + v^2} dv$$

$$= -\frac{a^2}{2g} \ln(v^2 + a^2) + c$$

when $x=0, v=u$

$$\Rightarrow c = \frac{a^2}{2g} \ln(u^2 + a^2)$$

$$x = -\frac{a^2}{2g} \ln(v^2 + a^2) + \frac{a^2}{2g} \ln(u^2 + a^2)$$

$$x = -\frac{a^2}{2g} [\ln a^2 - \ln(u^2 + a^2)]$$

$$= \frac{a^2}{2g} \ln \frac{u^2 + a^2}{a^2}$$

$$= \frac{a^2}{2g} \ln \left(1 + \frac{u^2}{a^2} \right)$$

b) $x = \cos^3 \theta \quad \frac{dx}{d\theta} = -3\cos^2 \theta \cdot \sin \theta$

$$y = \sin^3 \theta \quad \frac{dy}{d\theta} = 3\sin^2 \theta \cdot \cos \theta$$

$$\frac{dy}{dx} = \frac{3\sin^2 \theta \cos \theta}{-3\cos^2 \theta \sin \theta}$$

$$= -\tan \theta$$

Normal = cot θ at P.

Eqn of normal:

$$y - \sin^3 \theta = \frac{\cos \theta}{\sin \theta} (x - \cos^3 \theta)$$

$$\sin^4 \theta y - \sin^4 \theta = \cos^4 \theta x - \cos^4 \theta$$

$$x \cos^4 \theta - y \sin^4 \theta = \cos^4 \theta - \sin^4 \theta$$

$$= (\cos^2 \theta - \sin^2 \theta)$$

$$= \cos 2\theta$$

(i) $y_A = 0, x_A = \frac{\cos 2\theta}{\cos^4 \theta}$

$$x_B = 0, y_B = -\frac{\cos 2\theta}{\sin^4 \theta}$$

$$AB = \sqrt{\frac{\cos^2 2\theta}{\cos^2 \theta} + \frac{\cos^2 2\theta}{\sin^2 \theta}}$$

$$= \cos 2\theta \sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}}$$

$$= \frac{\cos 2\theta}{\sin \theta \cos \theta}$$

$$= \frac{\cos 2\theta}{\frac{1}{2} \sin 2\theta}$$

$$= 2 \cot 2\theta$$

Q5a) $P(x) = ax^3 + bx^2 + d = 0$

$P'(x) = 3ax^2 + 2bx$

Double root so root of $P(x)$ also root of $P'(x)$

$3ax^2 + 2bx = 0$

$x(3ax + 2b) = 0$

$x = 0$ or $x = -\frac{2b}{3a}$

$x = 0$ is not a root of $P(x)$

$\therefore x = -\frac{2b}{3a}$ is

$\therefore P\left(-\frac{2b}{3a}\right) = a \cdot \left(-\frac{2b}{3a}\right)^3 + b \cdot \left(-\frac{2b}{3a}\right)^2 + d = 0$

$-\frac{8ab^3}{27a^3} + \frac{4b^3}{9a^2} + d = 0$

$\times 27a^2$

$-8b^3 + 12b^3 + 27a^2d = 0$

$27a^2d + 4b^3 = 0$

b) $P(x) = \frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x + C = 0$

$P'(x) = x^3 - x^2 - 4x + 4$

$= x^2(x-1) - 4(x-1)$

$= (x-1)(x-2)(x+2)$

tps when $P'(x) = 0$

$x = \pm 2$ or 1

$P''(x) = 3x^2 - 2x - 4$

$P''(2) > 0$

$P''(-2) > 0$

$P''(1) < 0$

min tps at $x = \pm 2$

If $P(x)$ has no real roots

then $P(\pm 2) > 0$

$\frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x + C > 0$

$\therefore C > -1\frac{1}{3}$

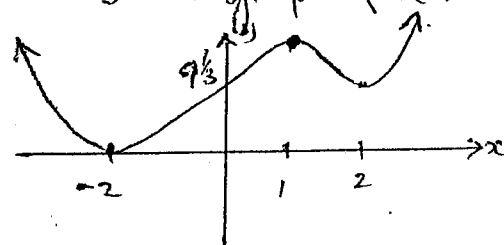
and

$\frac{1}{4}(-2)^4 - \frac{1}{3}(-2)^3 - 2(-2)^2 + 4(-2) + C > 0$

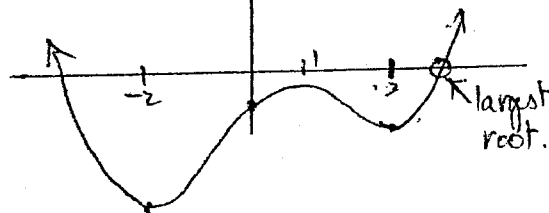
$\therefore C > 9\frac{1}{3}$

$\therefore C > 9\frac{1}{3}$

(ii) If $C = 9\frac{1}{3}$ the graph of $P(x)$ is



If $C = -2$



$P(2) < 0, P(3) = 3.25 > 0$

Take $x_0 = 2.5$

$x_1 = 2.5 - \frac{P(2.5)}{P'(2.5)}$

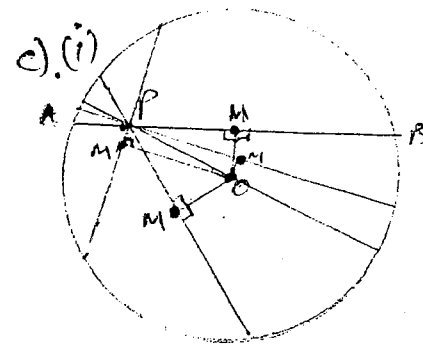
$= 2.5 - \frac{0.0572916}{3.375}$

$= 2.483$

$\therefore \text{largest root} = 2.483 \text{ (3dp)}$

$= \pi \left[\frac{\pi}{2} - 4C + C^2 \pi \right] \text{ units}^3$

(iii) min area

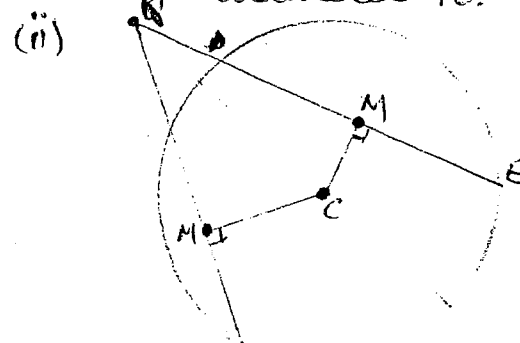


The locus of M is the circle with diameter OP.

The mid pt of any chord AB is the foot of the perpendicular from O.

$\therefore \angle OMA = \angle OMP = 90^\circ$

$\therefore$ Since angle in semi circle is 90° M lies on the circle with diameter PO.



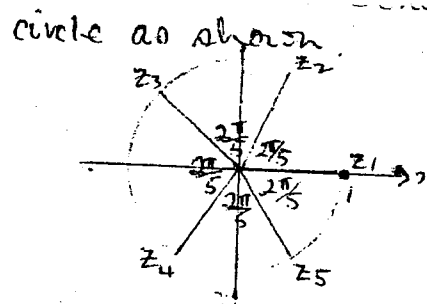
The locus of M is the arc of the circle with diameter PC which lies inside the circle centre C.

Max ht when $v = 0$

$x = -\frac{a^2}{2g} \left[\ln a^2 - \ln(u^2 + a^2) \right]$

$2I = \frac{2}{\pi} \ln 2$

$I = \frac{1}{\pi} \ln 2$



(i) $z_1 = 1$

$z_2 = \text{cis } \frac{2\pi}{5} = \omega$

$z_3 = \text{cis } \frac{4\pi}{5} = (\text{cis } \frac{2\pi}{5})^2 = \omega^2$

$z_4 = \text{cis } \frac{6\pi}{5} = (\text{cis } \frac{2\pi}{5})^3 = \omega^3$

$z_5 = \text{cis } \frac{8\pi}{5} = (\text{cis } \frac{2\pi}{5})^4 = \omega^4$

using de Moivre's theorem.

$1 + \omega + \omega^2 + \omega^3 + \omega^4 = \text{sum of roots of } z^5 - 1 = 0$
 $= -\frac{0}{1}$
 $= 0$

(iii)

The eqn with roots z_1 and z_2 is

$z^2 - (z_1 + z_2)z + z_1 z_2 = 0$

$z^2 - (\omega + \omega^4 + \omega^2 + \omega^3)z + (\omega^5 \omega^3)(\omega^2 + \omega^3) = 0$

$z^2 - (-1)z + \omega^3 + \omega^4 + \omega^6 + \omega^7 = 0$ from (ii)

$z^2 + z + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ since $\omega^6 = \omega^5 \omega = \omega$

$z^2 + z - 1 = 0$

$\omega^7 = \omega^5 \omega^2 = \omega^2$

$\cos(x+y) = \cos x \cos y - \sin x \sin y$

subtracting

$\cos(x-y) - \cos(x+y) = 2 \sin x \sin y$

$\sin x \sin y = \frac{1}{2} (\cos(x-y) - \cos(x+y))$
 $= \frac{1}{2} (\cos P - \cos Q)$

where $P = x-y$, $Q = x+y$

(ii) Prove $\sin x + \sin 3x + \sin 5x + \dots + \sin(2n-1)x = \frac{\sin^2 nx}{\sin x}$
 x by a line

Prove $\sin x \sin x + \sin 3x \sin x + \sin 5x \sin x + \dots + \sin(2n-1)x \sin x = \sin^2 nx$

LHS = $\frac{1}{2} (\cos 0 - \cos 2x) + \frac{1}{2} (\cos 2x - \cos 4x) + \dots + \frac{1}{2} (\cos(2n-2)x - \cos 2nx)$

$= \frac{1}{2} (1 - \cos 2nx)$

$= \frac{1}{2} (1 - (1 - 2\sin^2 nx))$

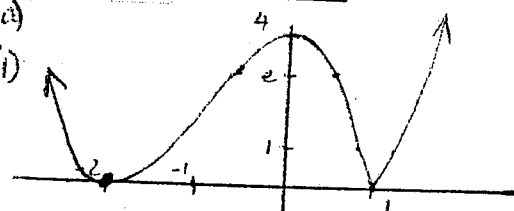
$= \frac{1}{2} \cdot 2\sin^2 nx$

$= \sin^2 nx$

$= \text{RHS}$

Q7a)

i)



(ii) If $y = 2\ln|x+2| + \ln|x-1| = m$

then $\frac{dy}{dx} = 0$

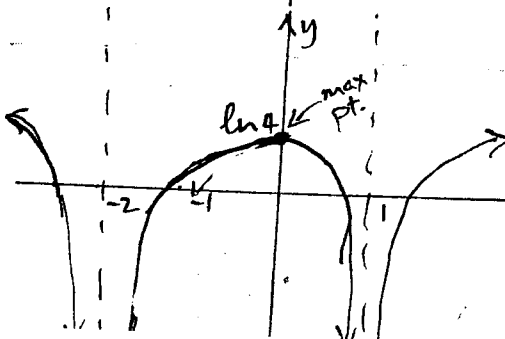
$y = \ln|x+2|^2 |x-1|$

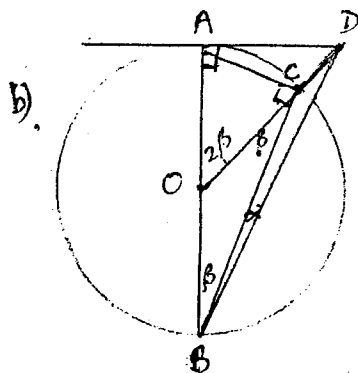
$= \ln|x^2 + 3x^2 - 4|$ as sketched.

Gradient of $y = \ln|x^2 + 3x^2 - 4|$

$= 0$ when

$y = m = \ln 4$





(i) Prove $\tan(\alpha + \beta) = \frac{1}{2} \tan 2\beta$.

$\angle AOC = 90^\circ$ (L between radii & tan).

$\angle ACD = 90^\circ$ (L in semi circle).

$$\tan(\alpha + \beta) = \frac{AD}{AB} = \frac{AD}{2AO}$$

$\angle AOC = 2\beta$ (ext L of isos Δ , radii $OB = OC$).

$$\frac{1}{2} \tan 2\beta = \frac{AD}{2AO}$$

$$\therefore \tan(\alpha + \beta) = \frac{1}{2} \tan 2\beta$$

(ii) Prove $\tan \alpha = \tan^3 \beta$.

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\frac{1}{2} \tan 2\beta = \frac{1}{2} \cdot \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

$$\therefore \frac{\tan \beta}{1 - \tan^2 \beta} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \text{ from (i)}$$

$$\tan \beta - \tan \alpha \tan^2 \beta = \tan \alpha + \tan \beta - \tan^2 \beta \tan \alpha - \tan^3 \beta$$

$$\tan^3 \beta = \tan \alpha$$

(iii) If $AD = AB$.

$\alpha + \beta = 45^\circ$ (L sum in isos rt L'd Δ).

$$\tan(\alpha + \beta) = \frac{1}{2} \tan 2\beta = 1$$

$$\tan 2\beta = 2$$

The roots lie on the unit circle as shown.

$$Q8a) \quad y = mx + c \quad \text{--- (1)}$$

$$(i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (2)}$$

Intersect when

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$(a^2 m^2 + b^2)x^2 + 2mca^2 x + (a^2 c^2 - a^2 b^2) = 0$$

If line is a tangent $\Delta = 0$

$$4m^2 c^2 a^4 - 4(a^2 m^2 + b^2)(a^2 c^2 - a^2 b^2) = 0$$

$$m^2 c^2 a^4 - a^4 m^2 c^2 + a^4 b^2 m^2 - a^2 b^2 c^2 + a^2 b^4 = 0$$

$$c^2 = \frac{a^4 b^2 m^2 + a^2 b^4}{a^4 b^2}$$

Prove $\frac{1}{a^2 m^2 + b^2}$
(ii) Tangents from (3, 4) to $\frac{x^2}{16} + \frac{y^2}{9} = 1$
are at rt Ls.

$$a = 4, b = 3$$

Tangents are $y = mx + c$

$$= mx \pm \sqrt{a^2 m^2 + b^2} \text{ from (i)}$$

$$= mx \pm \sqrt{16m^2 + 9}$$

These pass through (3, 4)

$$\therefore 4 = 3m \pm \sqrt{16m^2 + 9}$$

$$(4 - 3m)^2 = 16m^2 + 9$$

$$16 - 24m + 9m^2 = 16m^2 + 9$$

$$7m^2 + 24m - 7 = 0$$

Two values of m satisfy this, m_1 and m_2
and $m_1 m_2 =$ product of roots

$$= \frac{-7}{7}$$

$$= -1$$

$$(i) \cos(x-y) = \cos x \cos y + \sin x \sin y$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

subtraction

$$\text{Let } x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$x = \pm 1, \theta = \pm \frac{\pi}{2}$$

$$I_{2n} = \int_{-\pi/2}^{\pi/2} \cos^{2n} \theta \cos \theta d\theta$$

$$u = \cos^{2n} \theta$$

$$du = 2n \cos^{2n-1} \theta \cdot -\sin \theta d\theta$$

$$dv = \cos \theta d\theta$$

$$v = \sin \theta$$

$$I_{2n} = \left[\sin \theta \cos^{2n} \theta \right]_{-\pi/2}^{\pi/2} + 2n \int_{-\pi/2}^{\pi/2} \sin^2 \theta \cos^{2n-1} \theta d\theta$$

$$= 0 + 2n \int_{-\pi/2}^{\pi/2} (1 - \cos^2 \theta) \cos^{2n-1} \theta d\theta$$

$$= 2n \int_{-\pi/2}^{\pi/2} \cos^{2n-1} \theta - \cos^{2n+1} \theta d\theta$$

$$= 2n I_{2n-2} - 2n I_{2n}$$

$$(2n+1) I_{2n} = 2n I_{2n-2}$$

$$I_{2n} = \frac{2n}{2n+1} I_{2n-2}$$

$$= \frac{6}{7} I_4$$

$$I_6 = \frac{6}{7} \cdot \frac{4}{5} I_2$$

$$= \frac{6^2}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} I_0$$

$$= \frac{16}{35} \int_{-1}^1 (1-x^2)^2 dx$$

$$= \frac{16}{35} \left[x \right]_{-1}^1$$

$$= \frac{32}{35}$$

$$I_{2n} = \frac{2n(2n-2)(2n-4) \dots 4 \cdot 2}{(2n+1)(2n-1)(2n-3) \dots 5 \cdot 3} I_0 \times \frac{2n(2n-2)(2n-4) \dots 4 \cdot 2}{2n(2n-2)(2n-4) \dots 4 \cdot 2}$$

$$= \frac{[2n \cdot 2(n-1) \cdot 2(n-2) \cdot 2(n-3) \dots 2 \times 2 \times 2]^2}{(2n+1)!} I_0$$

$$= \frac{(2^n n!)^2}{(2n+1)!} \cdot 2$$

$$= \frac{2^{2n+1} (n!)^2}{(2n+1)!}$$